

LO4: Digital Maturity Diagnostics & Roadmaps

First: What is Digital Maturity?

Digital maturity refers to how effectively an organisation uses **digital technology, data, processes, people and organisational culture** to achieve its business goals.

It is **not simply about how much technology a company owns**.

For example:

Company A has 100 computers but still uses paper forms and makes decisions based on assumptions.

Company B has fewer technologies but uses cloud systems, data analytics, automation and digital processes effectively.

Company B may have higher digital maturity.

Four important pillars:

1. **Digital Culture & Leadership**
2. **Technology Stack Architecture**
3. **Data & Analytics Capabilities**
4. **Operations & Process Automation**

1. Digital Culture & Leadership

What does it mean?

This refers to whether the organisation's **leaders and employees are willing and able to adopt digital ways of working**.

It includes:

- Willingness to innovate
- Employee digital skills
- Leadership support
- Data-driven decision-making
- Experimentation
- Ability to adapt to change

Real-world example: DBS Bank



A bank cannot become digitally mature simply by creating a mobile banking app.

It also needs employees and managers who understand:

- Digital customer behaviour
- Data
- Cybersecurity
- AI
- Agile development
- Digital products

Example scenario

Imagine DBS wants to introduce an AI-powered financial assistant.

A **low-maturity organisation** might say:

"We bought an AI system. IT will take care of it."

A **more digitally mature organisation** asks:

"How can AI improve the customer experience, and how should our employees, processes and policies change to support it?"

Key lesson

Digital transformation is a people and culture transformation, not just a technology project.

2. Technology Stack Architecture

What does it mean?

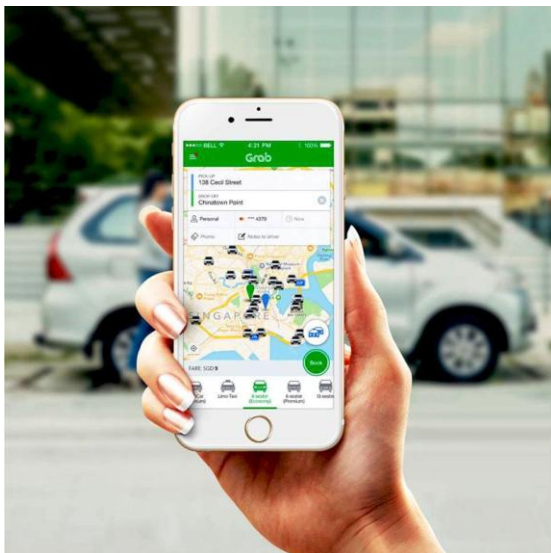
This refers to the **technology foundation that allows different digital systems to work together.**

The slide specifically mentions:

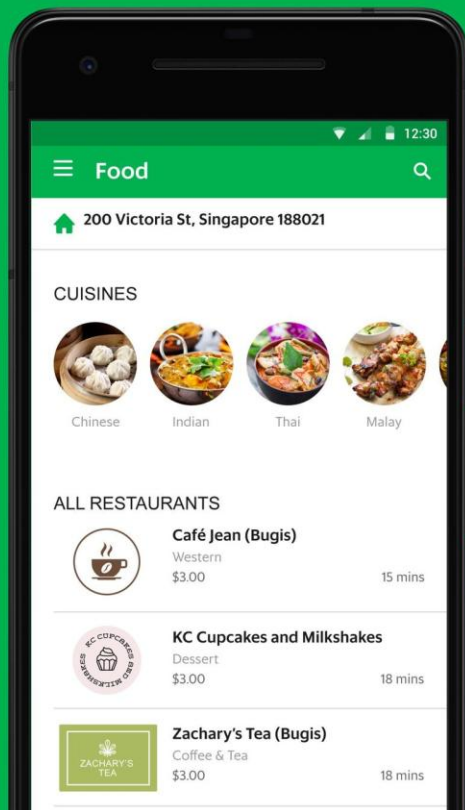
- Cloud-native systems
- API readiness
- Security compliance

Think of it as the organisation's **digital infrastructure and plumbing.**

Real-world example: Grab



**Browse all kinds of
food near you!**



A platform such as Grab provides multiple services through digital technology.

For example:

Ride booking

Food delivery

Payments

Customer accounts

Driver applications

Restaurant applications

These services need to communicate with each other.

For example:

Customer places food order

↓

Restaurant receives order

↓

Payment is processed

↓

Delivery partner is assigned

↓

Customer tracks delivery

↓

Customer receives notification

This requires an integrated technology architecture.

Low maturity

Systems are isolated:

POS ❌ CRM ❌ Website ❌ Mobile App

Employees manually transfer information between systems.

High maturity

Systems communicate through APIs:

POS ↔ CRM ↔ Mobile App ↔ Payment ↔ Delivery ↔ Analytics

Key lesson

Digital maturity requires technology systems that can communicate, scale and operate securely—not just individual applications.

3. Data & Analytics Capabilities

What does it mean?

A digitally mature organisation doesn't simply **collect data**.

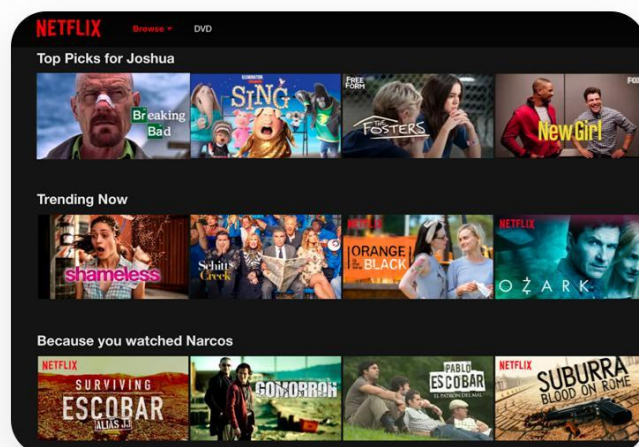
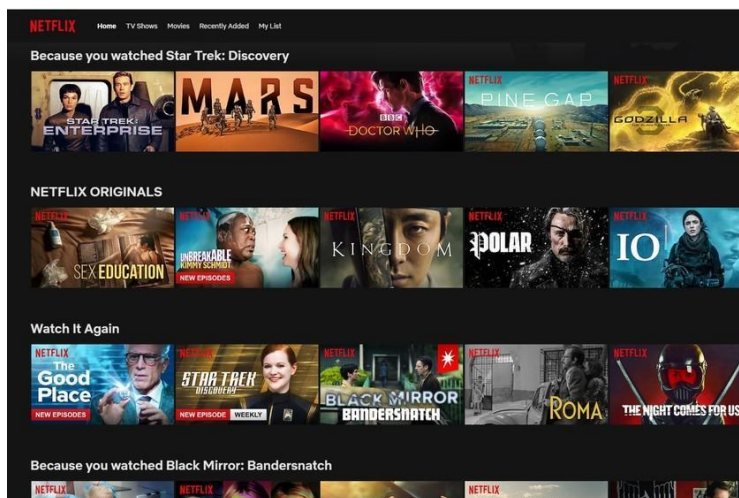
It can turn data into:

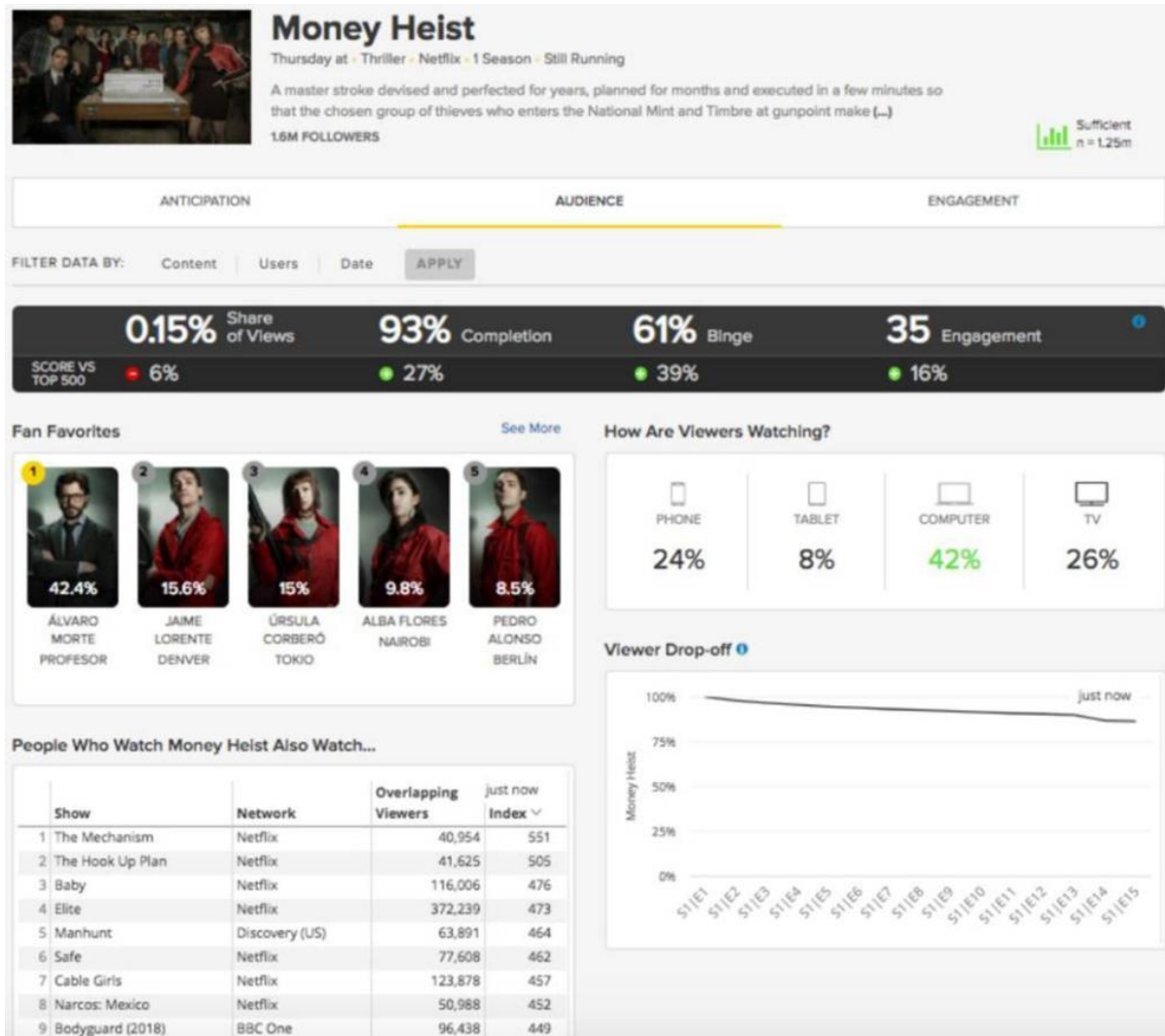
Information → Insights → Decisions → Action

The slide mentions:

- Real-time telemetry
- Predictive insights
- Unified data models

Real-world example: Netflix





Netflix can use viewing behaviour to understand customer preferences.

For example:

Customer watches three Korean dramas.

The system analyses viewing behaviour.

↓

It identifies patterns.

↓

It recommends other relevant content.

↓

Customer discovers another programme.

↓

Customer continues using Netflix.

Low data maturity

The company might simply collect:

Customer name + email + purchase history

but doesn't use the information effectively.

High data maturity

The organisation combines:

Customer data + behavioural data + transaction data + operational data

and uses analytics/AI to make decisions.

Key lesson

Data becomes valuable when an organisation can use it to make better decisions and improve customer experiences.

4. Operations & Process Automation

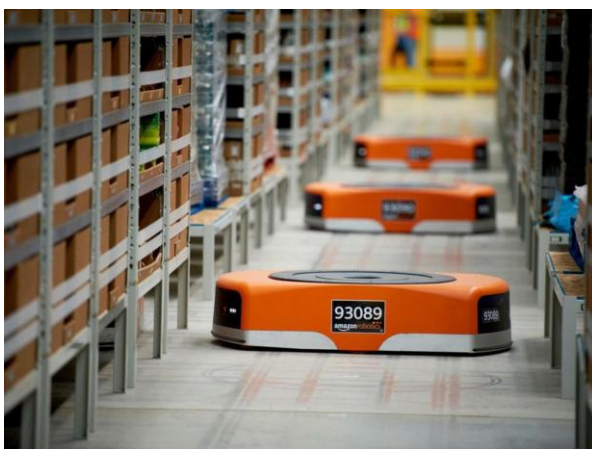
What does it mean?

This refers to using technology to **automate repetitive, manual or rule-based processes**.

Examples include:

- Automated invoices
- Chatbots
- Automatic notifications
- Workflow automation
- Robotic Process Automation (RPA)
- Automated testing and deployment
- Automated customer onboarding

Real-world example: Amazon





Amazon uses extensive technology and automation within its fulfillment operations.

Imagine a customer orders:

Laptop

The digital system can trigger a series of processes:

Order received

↓

Inventory checked

↓

Warehouse identifies item

↓

Item picked

↓

Packed

↓

Shipping label generated

↓

Delivery process begins

Much of this workflow can be digitally coordinated and automated.

Low maturity

Customer order → Employee prints order → Employee checks inventory → Employee manually enters information → Employee contacts delivery company

Higher maturity

Customer order → Integrated system → Automated inventory → Automated workflow → Delivery tracking

Key lesson

Automation reduces manual work, errors and processing time while allowing employees to focus on higher-value activities.